MULTI-AGENT AI FOR GAME DEVELOPMENT

BREACHPOINT

GDD — VERTICAL SLICE (OPTION C)

ASSIGNMENT	Capstone — 6-Week Vertical Slice
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ENGINE	Unreal Engine 5.8 — native C++ / GAS

1. Executive Summary

1.1 Concept

Breachpoint is a **4v4 team arena first-person shooter** built on the Halo sandbox model. Recharging **shields over finite health** give every fight a break-off point. A **two-weapon carry limit** and one contested power weapon make the map itself an objective. The **golden triangle** — shoot, grenade, melee — is available in every engagement, and a **Grappleshot** turns a three-level arena into a traversal playground: pull to a ledge, yank a rocket off the floor, close on a reloading enemy.

Bots fill every empty slot and fight through the same input path a human uses, so a match always starts.

This document specifies a **vertical slice**: the complete Halo feel at narrow breadth. Everything cut is listed in §6 with its Phase-2 restore path — nothing is silently dropped.

1.2 Win and Loss Conditions

- **Win:** first team to **25 kills**, or the higher team score when the **8:00** timer expires.
- **Tiebreak:** sudden death — no respawns, first kill wins, **60-second cap**; if uncontested, higher team damage dealt wins.
- **Loss:** any other scoreline at the timer.

1.3 Mode

Mode	Players	Description
Team Slayer	4v4 (humans + bots in any mix)	The only shipped mode. Bots fill every unfilled slot.

Solo play is Team Slayer with seven bots. **PvE is not a separate game** — it is the same code with the roster changed.

1.4 Platform, Format, and Production Reality

- **Engine:** UE 5.8, **pure native C++**, no Blueprint runtime logic. Combat, weapons, shields, and Grappleshot all on GAS.
- **Foundation:** stock **First Person template** (character, camera, Enhanced Input). Everything above it is authored in-house.
- **Networking:** server-authoritative, **Steam listen server** (host + invite), built behind `IOSServerLifecycle` so the dedicated-server/GameLift migration is a swap, not a rewrite.
- **Audio:** engine-native **MetaSounds** via GameplayCues.
- **Art:** sourced marketplace/free environment kit, weapon models, FPS arms + weapon animation sets, VFX. **Art is sourced; code is authored.**
- **Team & timeline:** one principal engineer / technical director plus the 12-agent crew, **6 weeks** (see §7 for the honest estimate).
- **Exit criterion:** playable **Steam demo depot**.

1.5 Elevator Pitch

Halo's sandbox — shields, two weapons, grenades, and a grappling hook — as a 4v4 arena shooter, written from scratch in C++ by one engineer and an AI crew in six weeks.

2. Game Mechanics (Player-Facing Actions and Loop)

2.1 The Match Loop

1. **Spawn** with the standard loadout: **Assault Rifle + Magnum**, 2 frag grenades, Grappleshot (recharging).
2. **Fight** — every engagement mixes the triangle: strip shields with fire, throw a grenade to deny the retreat, close with melee to finish.
3. **Contest the rocket** — the **Rocket Launcher** respawns on a fixed **90-second timer** at the arena's contested mid-level, with a countdown visible to both teams. It is the reason teams fight over a place instead of wandering.
4. **Die** → **respawn** in 5 seconds at the spawn point scored farthest from enemy pressure.
5. **Score or clock** — first to 25 or highest at 8:00 → carnage report → rematch.

2.2 The Golden Triangle — the non-negotiable core

All three are always available, and none alone wins an engagement:

Verb	Role	Detail
Shoot	Ranged damage	Two carried weapons; swap 0.4 s
Grenade	Area denial / flush	2 frags, physics-thrown, bounces off geometry
Melee	Finisher	70 dmg at contact; **rear melee = instant kill**

The rhythm — *shoot to break shields* → *grenade to cut the retreat* → *melee to finish* — **is** the game. If this is not fun by end of Week 2, the project stops (§7.2).

2.3 Shields Over Health — the signature system

Two damage layers, both GAS attributes on the PlayerState-owned ASC:

- **Shields (100)** — recharge at 60/s beginning **2.5 s** after the last damage taken. Distinct audio + visual cue on break and on recharge.
- **Health (100)** — **never regenerates.** The only reset is death.

Consequences that define the pacing: fights have a natural break-off point; a wounded player is a hunted player; re-engaging is a real decision, not a reflex. **Built first (Week 1)** because every other number is tuned against it.

2.4 Weapons — three, each a distinct decision

Weapon	Source	Type	Role
Assault Rifle	Spawn	Hitscan, 600 RPM	Spray; the shield-stripper you always have
Magnum	Spawn	Hitscan, ×2 headshot	Precision; the finisher on a broken shield
Rocket Launcher	**Map pickup, 90 s timer**	Projectile, AoE	The power weapon; map control; the demo's spectacle

Three weapons is the **floor** that still produces real sandbox decisions, because the two-weapon limit makes the rocket a *trade*: pick it up and you drop your AR or your Magnum — losing either your shield-stripping or your finishing tool. That trade is the sandbox.

*(The rocket also amortizes: it shares the projectile + radial-damage implementation with grenades, so weapon #3 costs a fraction of weapons

#1–2.)*

2.5 Grappleshot — the traversal pillar

A GAS ability on a **20-second cooldown**, 20 m range, three uses in one:

- **Pull to geometry** — yank yourself to a ledge (verticality, escape, flank).
- **Pull a weapon** — snatch the floor rocket from range. This single interaction makes the power-weapon contest dramatic.
- **Pull an enemy** — close distance and set up the melee finisher.

This is the concept's signature verb and the reason the map is vertical. It is also the **most netcode-sensitive system in the build** — a predicted movement ability with an attach point — and is therefore specced as a **netcode-builder** packet with a mandatory **critic REFUTER pass** before landing (§4.4).

2.6 The Map — one arena, three levels

A single symmetric arena with three connected elevations:

- **Lower** — enclosed, close-quarters, AR/melee territory
- **Mid** — the contested band; **Rocket Launcher spawn** at its center
- **Upper** — catwalks and ledges reachable **primarily by Grappleshot** (the pillar must be load-bearing for map access, not decorative)

Fixed constraints for the Arena Architect agent:

- 2 team spawn zones + 4 scored neutral spawn points (anti-camp)
- **Sightline cap 35 m** — long enough for Magnum duels, short enough to stay readable
- Every upper-level position reachable by at least **two** grapple points (no single-point chokes)
- Rocket spawn visible from all three levels (the contest must be legible)

2.7 Information Without Radar — a named design consequence

The full concept ships a motion tracker; **the slice cuts it** (§6). That removes the player's primary awareness tool, so the slice replaces it deliberately rather than accidentally:

- **Footstep and weapon audio are the information system.** MetaSounds spatialization is a *gameplay* requirement here, not polish — distinct, directional footsteps and clearly-audible reloads carry the awareness load radar would have.
- **Map readability compensates:** the 35 m sightline cap and open three-level sightlines mean threats are usually visible before they are lethal.

This is a real trade and it is named as one: without radar the slice plays slightly more twitchy and less tactical than full Halo. It is the right cut for six weeks; it is the **first system restored** in Phase 2.

2.8 Bots — one brain, dialed

Bots occupy every unfilled slot. The slice ships **one authored behavior profile** scaled by a difficulty multiplier — giving three player-facing difficulties without authoring three behavior trees:

Setting	Accuracy	Reaction	Grenade use	Behavior
Recruit	25%	500 ms	Rare	Same StateTree, dulled
Marine *(default)*	45%	320 ms	Situational	Baseline profile
Veteran	65%	220 ms	Tactical	Same StateTree, sharpened

All settings drive the **same** StateTree and EQS queries — only

`DT_BotTuning` scalars change. Bots activate abilities through the same input path a human uses: no aimbot hooks, no privileged state, no side-channel damage. This is what makes bot-vs-bot soak tests exercise the real combat code (§4.5).

2.9 What the Player Sees (HUD)

- **Top-left:** shield bar over health bar — the two-layer read is the most important element on screen.
- **Bottom-right:** current weapon + ammo + spare mags; second weapon ghosted beneath.
- **Bottom-left:** grenade count; **Grappleshot cooldown ring**.
- **Center:** reticle with **distinct hit markers for shield hits vs. flesh hits** — the sandbox is unreadable without this distinction.
- **Top-center:** team score, match timer, **rocket respawn countdown**.
- **Feed:** killfeed + medals (Double Kill, Killing Spree, **Grapple Kill**, Rocket multi-kill) — **canned, instant, deterministic**.
- **Match end:** carnage report (K/D/A, accuracy, medals) + one Spotter coach line + rematch prompt.

3. AI Architecture (What Each Agent Does, Through Its Effect on Gameplay)

The slice's AI story is deliberately **bots-first**: real game AI is a stronger showcase than an LLM narrator, and it is the part a player actually experiences. The LLM stays out of the simulation entirely.

3.1 In-Game AI — StateTree + EQS (no LLM)

Bot Brain — StateTree. States: `Seek → Engage → Flush → Reposition → Retreat → ContestRocket`. StateTree is UE 5.8's production-ready default and merges behavior-tree selectors with state-machine transitions — the correct shape for combat AI. Transitions are driven by shield state, ammo, threat count, and the rocket timer.

Cover and positioning — EQS. Environment Query System scores candidate positions by cover from the current threat, distance to the rocket spawn, and flank angle. This is what makes a bot *read* as playing rather than pathing.

Explicitly not used: MassAI (experimental — not shippable in a capstone) and **Learning Agents** (reinforcement learning is a research project, not a six-week feature). StateTree + EQS is the current, professional, shippable choice.

Determinism: bot decisions are a pure function of (tuning row, match seed, observed events); reaction jitter is seeded once per match, so QA soaks reproduce exactly and any disputed result is replayable.

3.2 The Dev-Time Crew (Claude + Unreal MCP)

The formulated 12-agent crew carries over unchanged; only its products change. Every output is reviewable data (JSON/DataTables), refuted by the critic, landed by a builder. **Agents never commit directly.**

Agent	Product for the slice
Arena Architect	`arena_manifest.json` — three-level layout, grapple points (≥2 per upper position), scored spawns, rocket node, 35 m sightline cap
Bot Trainer	`DT_BotTuning` — the baseline profile + the three difficulty scalars
Weapon Curator	`DT_Weapons` rows (AR/Magnum/Rocket) with TTK analysis against the shield+health model
Combat QA	Nightly bot-vs-bot soaks: stuck navmesh, unreachable grapple points, spawn-kill loops, TTK regressions
Balance Analyst	Weapon win-rate/TTK bands; rocket-control vs. match-win correlation

3.3 The Runtime Agent — Spotter

Spotter Agent — the only model call in the shipped game, and never load-bearing.

- **Role:** one telemetry-grounded coach line per human player at match end; optional color commentary appended to notable killfeed events (sprees, rocket multi-kills, grapple kills).
- **Gameplay effect:** every player leaves with one specific, *earned* correction — *"You lost 6 fights below 40% shields — break off and let them recharge."*
- **Output:** `{coach_line (≤ 30 words, references ≥ 1 telemetry stat)}`; per event `{spotter_line (≤ 18 words) | null}`.
- **Never load-bearing:** medals and killfeed are **canned and instant** — Halo's announcer is deterministic by design, which is correct, not a limitation. Spotter lines only *append*. Host-side, async, ≤ 12 calls/match, 3 s timeout, canned-line DataTable fallback shipped in the build. **No connectivity ⇒ the game is identical minus flavor.**

3.4 Trust Boundaries

- **No LLM in the simulation path.** Its only output is display strings.
- **Bots are server-side only**; clients see a replicated player.
- **The API key never leaves the host**; clients receive strings.
- **Hidden information stays hidden at the replication layer**, not the render layer — enemy transforms use the tightest replication conditions and relevancy culling available.

4. Technical Strategy

4.1 Stack — authored, ported, sourced

Layer	Origin
First-person character, camera, Enhanced Input	**UE 5.8 FPS template**
GAS core (PlayerState ASC, input buffer, prediction)	**Ported from OnSight** — own code
Steam sessions	**Ported from OnSight** (`OSSessionsSubsystem`)
Shields/health, weapons, grenades, melee, **Grappleshot**	**Authored** — GAS abilities + effects
Bot AI	**Authored** — StateTree + EQS
Team Slayer, scoring, respawn scoring, rocket timer	**Authored**
HUD + front end	**Authored** — CommonUI, C++ base classes + BP visuals
Audio	**Authored** — MetaSounds via GameplayCues
Server lifecycle abstraction	**Authored** — `IOSServerLifecycle`
Meshes, animations, environment kit, VFX	**Sourced**

4.2 Token Budget (runtime, per match)

Call	Model	Calls / match	Tokens in	Tokens out	Total
Spotter (events, batched ≥ 10 s)	Claude Haiku	≤ 12	600	60	≈ 7,900
Coach (match end, per human)	Claude Haiku	≤ 8	900	80	≈ 7,800
Total					≈ **15,700** ≈ **$0.013 / match**

Caps enforced server-side; every failure path falls back to canned lines.

4.3 Constraints (named, with the decision each caused)

1. **Constraint: six weeks, one principal, 100% authored gameplay code.** *Therefore* the sandbox ships at **three weapons and one map**, the motion tracker is cut, bots ship as one scaled profile, and vehicles were never in scope. The full sandbox is Phase 2 (§6).
2. **Constraint: no Lyra, no third-party gameplay code.** *Therefore* every system Lyra would have donated is authored — which is the honest source of this project's schedule risk (§7), and the reason the GAS core is **ported from our own shipped codebase** rather than rebuilt.
3. **Constraint: Grappleshot is a predicted movement ability.** Prediction is the highest-risk code an agent crew writes (the silent-and-confident class). *Therefore* it is a netcode-builder packet, ships with its own cheat-attempt test, and requires a critic REFUTER pass and a rung-4 Gauntlet run under 120 ms emulation before landing.
4. **Constraint: GameLift plugin support lags UE 5.8, and infra work in Week 1 kills capstones.** *Therefore* the slice ships on **listen server** behind `IOSServerLifecycle`, with the dedicated-server/ GameLift fleet as a **post-course swap** rather than a six-week dependency.
5. **Constraint: LLM latency (1–4 s).** *Therefore* medals and killfeed are canned and instant; Spotter is async and additive only.

4.4 Netcode Rules (slice-specific)

- **Server-authoritative hit registration.** Clients send fire intent; the server validates and applies damage. No client-declared kills.
- **Every** `Server` **RPC ships** `WithValidation` with a real `_Validate` body — range, rate, and possession checks. Empty `return true;` fails review.
- **Grapple prediction reconciles.** A rejected grapple must leave zero gameplay state — no position fork, no cooldown consumed, no cue residue.
- **Replicate the minimum.** `COND_OwnerOnly` for private state (ammo reserves, exact shield value); relevancy and net-cull tuned per class.
- **The attack ships with the feature.** Each new replicated surface lands with its own cheat-attempt test whose **rejection** is the acceptance criterion; the critic re-attacks independently.

4.5 Verification (the ladder)

Per `crew/docs/contracts/testing.md`: clean compile ⟶ headless Automation Specs (`Breachpoint.Sim.*` pins TTK, shield recharge, damage math; `Breachpoint.Bots.*` pins determinism: same seed + tuning row ⇒ identical action trace) ⟶ functional tests ⟶ **Gauntlet networked smoke** (server + 2 clients, assert in threes, then again under network emulation at `120 ms` lag / `5%` loss) ⟶ perf spot-checks. **A rung that cannot run is BLOCKED, never skipped.** Overnight bot-vs-bot soaks run nightly from Week 4.

5. Scope and Schedule

5.1 Shipped Scope

Team Slayer 4v4 (bots filling any slot, 3 difficulty settings); **1** three-level arena map; **3** weapons (AR, Magnum, Rocket on a 90 s timer); frag grenades; melee including rear-kill; **Grappleshot**; shields-over-health; scored respawns; CommonUI HUD + minimal front end; canned medals + killfeed; Spotter agent with canned fallback; MetaSounds combat and footstep audio; Steam listen server + demo depot. **Nothing else.**

5.2 Schedule — six weeks, each ending runnable

Wk	Deliverable	Gate
1	Ladder bootstrap (specs + Gauntlet skeleton); FPS template + module layout; **GAS core ported**; **shields/health**; AR firing; `IOSServerLifecycle` stub	Breaking a dummy's shields feels good
2	Magnum + two-weapon carry + swap; **grenades**; **melee + rear-kill**	■■ **THE GOLDEN TRIANGLE FUN TEST** — see §7.2
3	**Grappleshot** (netcode packet + REFUTER pass); map blockout from Arena Architect manifest	Traversal is fun; grapple is used offensively
4	Bots (StateTree + EQS + 3 scalars); Team Slayer scoring, teams, scored respawns; nightly soaks begin	■■ **GO / NO-GO** — a full 4v4 match vs. bots plays end-to-end
5	Rocket Launcher + 90 s timer; CommonUI HUD; Steam listen server + host/invite; MetaSounds cues	Two humans + six bots, online, full match
6	Balance pass (crew telemetry); polish; Steam demo depot; capstone presentation	**Shipped**

5.3 Cut Order (pre-declared — used, not improvised)

If behind at any gate, cut **in this order** and log it:

1. **Rocket Launcher** → two weapons; power-weapon contest becomes a map-control point without a reward *(costs the sandbox trade — cut only if Week 5 is at risk)*
2. **Bot difficulty settings** → single "Marine" profile
3. **Medals** → killfeed only
4. **Spotter agent** → canned coach lines only *(the fallback already ships, so this is a config flip)*
5. **Front-end menus** → direct-to-match + Steam overlay invite only

Never cut: shields-over-health, the golden triangle, Grappleshot. Those three *are* the game; cutting any of them means building a different, worse project.

5.4 First Restore If Ahead

1. **FFA Slayer** (same scoring code, team assignment off — near-free)
2. **Motion tracker** (the most-missed cut system, §2.7)
3. Plasma Rifle + the plasma/kinetic damage-type layer

6. What's Cut vs. Full Breachpoint — and the Phase-2 path

Nothing here is silently dropped. Each cut has a named restore path.

Cut from full concept	Why	Phase 2 restore
Motion tracker	0.5 w; awareness carried by audio + sightlines (§2.7)	First restore post-slice; needs server-computed contacts
Plasma Rifle + damage-type layer	With only kinetic weapons the damage table is trivial	Add with weapon #4; table already schema'd
Bot tiers 2–3 as distinct behaviors	One scaled profile delivers 3 difficulties	Author distinct StateTrees per tier
Dedicated server + GameLift fleet	1.5 w of infra that blocks gameplay work	**Interface already in place** (`IOSServerLifecycle`) — a swap, not a rewrite
FFA + Firefight modes	Mode variety is not the slice's thesis	Both are roster/team-assignment configs
Second map	Content, not systems	Arena Architect produces from the same manifest schema
Vehicles	Networked multi-occupant physics — the genre's biggest scope trap	Not planned; Grappleshot carries traversal permanently

7. Honest Schedule Assessment

7.1 The estimate

Re-derived line-by-line from the build inventory in

`../SCOPE-COMPARISON.md`, with the Option-C cuts applied:

	Weeks
Raw build inventory	≈ **8.4**
After crew compression (~25%, content and review only)	≈ **6.3**
Window	**6.0**

> **This fits six weeks only with near-zero slippage. It fits seven > comfortably.** The earlier ≈5.5–6 w figure in the comparison document > was optimistic; this is the corrected number.

The plan absorbs the gap three ways: the **pre-declared cut order** (§5.3) is expected to be *used*, not held in reserve; **art is sourced, never authored**; and the crew owns all content production (map, tuning, QA) so the principal's weeks go to systems only.

7.2 The two gates that matter

Week 2 — the Golden Triangle fun test. Shoot/grenade/melee against a stationary dummy and one scripted bot. If the loop is not fun here, **no amount of later content saves it** — and Concept A's shipped foundation is still available. This gate exists to make that switch cheap while it is still cheap.

Week 4 — go/no-go. A full 4v4 vs. bots must play end-to-end. If it does not, execute the cut order down to two weapons and one bot profile, and protect Weeks 5–6 for shipping. **Shipping a smaller slice beats shipping nothing** — that is the entire lesson of the scope analysis.

7.3 Risk register

Risk	Severity	Mitigation
Schedule (~6.3 w into 6 w)	**High**	Pre-declared cut order used actively; Week-4 go/no-go; art fully sourced
Grapple prediction under latency	**High**	netcode-builder packet; cheat-test ships with feature; critic REFUTER; rung-4 at 120 ms before landing
Bots read as robotic (Halo's AI is the bar)	Medium	EQS-driven cover; nightly soaks from Week 4; tuning is data, iterable to the last day
Sourced animations don't fit authored abilities	Medium	**Author ability timings to the acquired anim sets**, never the reverse — decide the anim pack in Week 1
No radar makes fights feel blind	Medium	Audio is a gameplay requirement (§2.7); sightline cap; restore radar first if ahead
Listen-server host advantage	Medium	Standing review item on every netcode packet; host and remote claims tested separately

Risk	Severity	Mitigation
StateTree/EQS learning curve	Low–Med	Excluded from estimates deliberately; Week 4 is the buffer-consuming week if it bites

8. Success Criteria

- A stranger installs the Steam demo, joins a 4v4 with bots, and gets a kill inside **two minutes**, unaided.
- Playtesters use the **Grappleshot offensively** — not just to travel — the sign the pillar landed.
- Playtesters ask for another match unprompted at the carnage report.
- Every map decision, bot value, and weapon number in the build traces to a **named agent output that a human reviewed**.

Appendix A — Combat Tuning (first pass)

Parameter	Value
Shields / Health	100 / 100
Shield recharge	60/s, begins 2.5 s after last damage
Health regeneration	none
Assault Rifle	8 dmg/shot, 600 RPM, 32 mag, hitscan
Magnum	22 dmg, ×2 headshot, 8 mag, hitscan
Rocket Launcher	120 dmg, radius 4 m, 2 shots, **90 s respawn**
Frag grenade	90 dmg centre, radius 5 m, 2 carried
Melee	70 dmg; **rear = instant kill**
Grappleshot	20 s cooldown, 20 m range
Weapon swap	0.4 s
Respawn	5 s, scored spawn
Match	25 kills or 8:00; sudden death 60 s cap
Map sightline cap	35 m

Appendix B — Bot Tuning Schema

`DT_BotTuning`: `profile_id`, `display_name`, `accuracy_pct`, `reaction_ms`, `burst_discipline`, `grenade_policy ∈ {rare, situational, tactical}`, `cover_preference (0..1)`, `push_threshold (enemy shield %)`, `rocket_contest (bool)`, `target_switch_bias`. **No profile may set `reaction_ms` below 200** (superhuman guard, enforced by the schema and asserted in `Breachpoint.Bots.*`).

Appendix C — Telemetry Schema

`FOS_MatchTelemetry` (per player, per match): kills, deaths, assists, accuracy per weapon, TTK distribution, shield-break→kill conversion, grenade kills, melee kills (front/rear), grapple uses and grapple kills, rocket holds and rocket kills, **fights lost below 40% shields**, medals, time alive. Consumed by: Spotter (coach lines), Balance Analyst (TTK/win bands), Combat QA (regression baselines).